

The Purification Paradox: Dissecting and Exploiting Generative Vulnerability Bands in Diffusion Trajectories

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Abstract

Denoising Diffusion Probabilistic Models (DDPMs) are powerful generative models, but their vulnerability to adversarial manipulation during reverse sampling remains poorly understood. Existing attacks typically perturb intermediate latent states or rely on heuristic timestep selection, often achieving high attack success at the cost of severe perceptual degradation. We analyse the reverse diffusion trajectory and identify a phenomenon that we term the Purification Paradox: timesteps with strong local sensitivity do not necessarily yield effective attacks, because subsequent denoising steps can suppress or absorb the perturbation. Motivated by this observation, we shift the attack surface from the latent state to the noise prediction space and introduce Unrestricted ϵ -Space Perturbation (U ϵ P). Evaluated within an intermediate Generative Vulnerability Band, U ϵ P achieves a stronger balance between attack success and perceptual fidelity than direct latent perturbation. The resulting perturbations also transfer effectively across unseen classifiers, reaching 71.4% ASR on MobileNetV2 while maintaining high visual fidelity (SSIM > 0.76).

1. Introduction

Denoising Diffusion Probabilistic Models (DDPMs) have become a central class of generative models because of their strong synthesis quality across a wide range of tasks [3, 7]. As these models are increasingly used in security sensitive settings, including synthetic media generation, digital forensics, and biometric systems, it becomes important to understand how adversarial perturbations interact with the reverse diffusion process. In particular, an attack that alters model output while preserving perceptual plausibility poses a direct threat to downstream decision systems.

Existing diffusion attacks often perturb intermediate latent states or optimise across a large portion of the reverse trajectory. Although such approaches can achieve high at-

tack success, they are often computationally expensive and can substantially degrade image quality. Prior work has mainly attacked diffusion models by perturbing latent states x_t [1, 11] or by selecting perturbation intervals heuristically, as in TAIGen [5]. While effective at inducing misclassification, these approaches provide limited control over the trade off between attack success and perceptual fidelity.

In this work, we study adversarial vulnerability along the reverse diffusion trajectory by shifting the attack surface from the latent state x_t to the noise prediction space ϵ_θ . This allows us to analyse how perturbations interact with the reverse dynamics of the model rather than only with the current latent representation. Our analysis reveals a phenomenon that we term the *Purification Paradox*: timesteps with high local sensitivity are not necessarily the most effective attack points, because subsequent denoising steps can suppress or absorb the injected perturbation.

Motivated by this observation, we introduce Unrestricted ϵ -Space Perturbation (U ϵ P), which applies iterative perturbation in the noise prediction space without step wise projection. We further identify an intermediate temporal region, termed the *Generative Vulnerability Band*, in which semantic structure has emerged while the trajectory remains sufficiently steerable for adversarial manipulation. Experiments show that this setting provides a stronger balance between attack success and perceptual fidelity than direct latent perturbation, while also producing strong transferability across unseen classifiers.

Our main contributions are as follows:

- We analyse adversarial vulnerability along the reverse diffusion trajectory and identify the *Purification Paradox*, showing that locally sensitive timesteps are not necessarily the most exploitable.
- We introduce Unrestricted ϵ -Space Perturbation (U ϵ P), an attack in the noise prediction space that enables stronger steering than bounded perturbation in the same space.
- We identify a *Generative Vulnerability Band* in the intermediate reverse trajectory, where attacks achieve a stronger balance between attack success and perceptual

fidelity.

- We show that perturbations generated in this regime transfer effectively across unseen classifiers while maintaining high visual fidelity.

2. Related Work

2.1. Diffusion Dynamics and Timestep Selection

Denoising Diffusion Probabilistic Models (DDPMs) generate data by reversing a forward Gaussian noise process [3]. This framework was later connected to continuous time formulations based on Stochastic Differential Equations (SDEs) and the probability flow ODE [7]. In practice, however, standard DDPM sampling is implemented as a discrete Markov chain of T reverse steps. For adversarial analysis, this makes it important to distinguish between local timestep sensitivity and the cumulative effect of perturbations on the final generated sample.

Evaluating attacks across all T timesteps is often uninformative: brute force gradient accumulation can achieve high attack success, but usually at high computational cost and with substantial perceptual degradation. A more informative setting is to examine an intermediate temporal region of the reverse process, where semantic structure is emerging and the trade off between attack effectiveness and perceptual fidelity becomes more meaningful.

2.2. Adversarial Attacks and the Discrete Assumption

Adversarial manipulation has extended from GANs to diffusion models. AdvDiff [1] demonstrated unrestricted adversarial generation through gradient based intervention during reverse sampling, while ACE [11] analysed vulnerabilities in Latent Diffusion Models (LDMs), identifying the latent state x_t as a primary attack surface.

Training free approaches such as TAIGen [5] introduced heuristic temporal gating by selecting specific mixing step intervals to improve transferability. However, most existing methods share a common limitation: they mainly treat the reverse process as a sequence of attackable latent states. Whether through direct perturbation of x_t or heuristic interval selection, prior work provides limited analysis of how adversarial effects propagate through the reverse trajectory, especially when perturbations are introduced in the noise prediction space ϵ_θ .

2.3. Adversarial Purification and Trajectory Defenses

Diffusion models have also been studied for their purification properties. DiffPure [4] showed that forward diffusing and then reversing an adversarial image can suppress high frequency perturbations and project corrupted inputs back toward the natural image manifold.

While prior work has primarily used purification as a defence mechanism [4], we consider it here as a constraint on adversarial manipulation within the reverse process itself. Rather than viewing diffusion only as an external purifier, we analyse how the remaining reverse steps can suppress or absorb perturbations introduced at different timesteps. This trajectory level perspective motivates the Purification Paradox and the shift from direct latent perturbation to attacks in the noise prediction space.

3. The Attack Surface: Formulating the Perturbation Mechanisms

To analyse adversarial vulnerability along the reverse diffusion trajectory, we first define the generative process and then identify the points at which adversarial perturbations can be introduced. Our goal is to distinguish direct perturbation of the latent state from perturbation of the predicted noise, and to study how these different interventions affect both attack success and perceptual fidelity.

3.1. Diffusion pipeline basics

We briefly summarise the Denoising Diffusion Probabilistic Model (DDPM) [3]. DDPM defines a discrete Markov forward process that gradually adds Gaussian noise to a clean sample x_0 , producing latent states $\{x_t\}_{t=0}^T$. The forward transition is given by

$$p(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t \mathbf{I}), \quad (1)$$

where $\{\beta_t\}_{t=1}^T$ is a predefined noise schedule. Let $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$. The reverse sampling step is parameterised by a neural network ϵ_θ , which predicts the noise component and updates the trajectory as

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(x_t, t) \right) + \sigma_t \mathbf{z}, \quad (2)$$

where $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})$ and σ_t controls the stochastic component of the reverse transition. The network ϵ_θ is trained to predict the injected Gaussian noise under the standard simplified DDPM objective [3].

In this formulation, x_t denotes the current latent state along the reverse diffusion trajectory, while $\epsilon_\theta(x_t, t)$ determines how that state is updated toward the data manifold. This distinction is central to our study. We consider attacks that either perturb the latent state directly or intervene through the predicted noise used in the reverse transition.

3.2. Defining the adversarial modes

Let $\mathcal{L}$ denote the untargeted adversarial loss computed from the reconstructed image and the classifier f_ϕ . We consider three attack modes that differ in where the perturbation is introduced and whether it is explicitly bounded at each optimisation step.

Mode 1: Latent perturbation (x_t mode) In the first mode, the intermediate latent state x_t is perturbed directly. In implementation, this is carried out using momentum iterative sign based updates with projection onto an ℓ_∞ ball around the current target latent state. At iteration k , this can be written abstractly as

$$x_t^{(k+1)} = \Pi_{\mathcal{B}_\infty(x_t^{(0)}, \epsilon/255)} \left(x_t^{(k)} + \alpha \text{sign}(m^{(k)}) \right), \quad (3)$$

where $\Pi_{\mathcal{B}_\infty(x_t^{(0)}, \epsilon/255)}(\cdot)$ denotes projection onto the ℓ_∞ ball of radius $\epsilon/255$ centred at the initial latent state, α is the step size, and $m^{(k)}$ is the accumulated momentum term. This mode directly alters the reverse trajectory at the latent level and typically provides the strongest control over the final output, although often at the cost of substantial perceptual degradation.

Mode 2: Bounded noise prediction perturbation (ϵ mode) In the second mode, the perturbation is applied to the predicted noise rather than to the latent state itself. The bounded variant uses the same momentum iterative optimisation, but applies projection after each update:

$$\hat{\epsilon}_\theta^{(k+1)} = \Pi_{\mathcal{B}_\infty(\epsilon_\theta^{(0)}, \epsilon/255)} \left(\hat{\epsilon}_\theta^{(k)} + \alpha \text{sign}(m_\epsilon^{(k)}) \right), \quad (4)$$

where $\hat{\epsilon}_\theta^{(0)} = \epsilon_\theta(x_t, t)$ is the unperturbed noise prediction and $m_\epsilon^{(k)}$ denotes the momentum term in the noise prediction space. This mode preserves explicit step wise boundedness and typically produces higher visual fidelity, but its attack strength is limited by repeated projection.

Mode 3: Unrestricted ϵ -Space Perturbation (U ϵ P) In the third mode, the perturbation is again applied in the noise prediction space, but without step wise projection:

$$\hat{\epsilon}_\theta^{(k+1)} = \hat{\epsilon}_\theta^{(k)} + \alpha \text{sign}(m_\epsilon^{(k)}). \quad (5)$$

Because no projection is imposed at each optimisation step, the perturbation can accumulate more persistently across iterations and across the attacked temporal window. This allows stronger steering of the reverse trajectory while often preserving substantially better image quality than direct perturbation of x_t .

Integration into the reverse diffusion transition Unlike standard classification attacks, perturbations introduced during the diffusion process must be propagated through the reverse sampling equations. For the two noise space modes, once the adversarially perturbed noise prediction $\hat{\epsilon}_\theta$ has been obtained, we first compute the corresponding clean estimate

Table 1. Baseline comparison over a full 100 step reverse trajectory. Direct perturbation of x_t yields the highest attack success but severely degrades perceptual quality. By contrast, U ϵ P maintains substantially higher fidelity while also achieving very high attack success.

Method	ASR (%) $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Clean / No Attack	17.2	23.31	0.77	0.022
100 step x_t attack	100	17.78	0.44	0.099
100 step ϵ attack	24.20	23.31	0.77	0.022
100 step U ϵ P	100	22.94	0.74	0.023

$$\hat{x}_0 = \frac{1}{\sqrt{\bar{\alpha}_t}} x_t - \sqrt{\frac{1 - \bar{\alpha}_t}{\bar{\alpha}_t}} \hat{\epsilon}_\theta. \quad (6)$$

We then substitute this estimate into the standard DDPM posterior mean

$$\tilde{\mu}_t = \frac{\sqrt{\bar{\alpha}_{t-1}} \beta_t}{1 - \bar{\alpha}_t} \hat{x}_0 + \frac{\sqrt{\bar{\alpha}_t} (1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} x_t, \quad (7)$$

and sample the next latent state as

$$x_{t-1} = \tilde{\mu}_t + \sigma_t \mathbf{z}.$$

This procedure ensures that the perturbed noise prediction is incorporated through the standard reverse DDPM update, so that the attack remains consistent with the model’s sampling mechanism.

3.3. Baselines and the fidelity bottleneck

Before analysing temporally localised vulnerabilities, we first establish empirical upper and lower bounds using attacks applied across the full 100 step reverse trajectory considered in our evaluation.

Table 1 reports the baseline results. Here, ASR is defined as the misclassification rate of the classifier on the reconstructed image with respect to the ground truth class label. Under this definition, the clean condition already has a non-zero ASR of 17.2%. Specifically, while the original images (without DDPM) achieve 96.20% accuracy (a 3.80% ASR), unconditional DDPMs inherently introduce a base reconstruction error and distribution shift. This naturally causes the downstream classifier to misclassify 17.2% of the unattacked clean reconstructions.

The comparison reveals a clear trade off. Directly perturbing x_t across all 100 steps produces near complete attack success, but with severe loss of perceptual quality. The bounded ϵ mode preserves image quality well, but provides only limited attack strength. U ϵ P provides a substantially better compromise: it reaches very high attack success while maintaining fidelity close to that of the unattacked reconstruction.

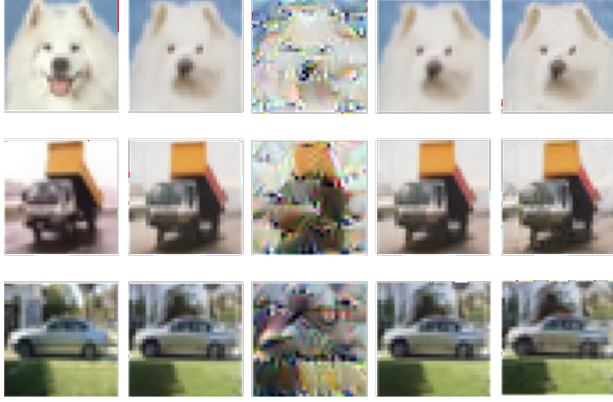


Figure 1. Visual comparison across attack modes. From left to right: real CIFAR 10 image, DDPM reconstruction without attack, x_t attack, bounded ϵ attack, and UeP attack. Direct perturbation of x_t introduces severe artefacts and structural degradation, whereas UeP alters classifier relevant content while preserving much of the perceptual structure.

These baseline results motivate the remainder of the paper. Although full trajectory attacks in the noise prediction space can achieve strong attack success with relatively good fidelity, attacking all 100 reverse steps is computationally expensive and does not reveal which temporal region is actually responsible for this vulnerability. This motivates a more focused temporal analysis of the reverse trajectory, with the aim of identifying when adversarial steering is most effective and how long it must be applied.

4. Temporal Dynamics and the Purification Paradox

Having established that a full 100 step attack in the noise prediction space is computationally expensive, we next investigate where along the reverse diffusion trajectory adversarial steering is most effective. A natural assumption is that vulnerability should concentrate at a few isolated timesteps, or that it can be captured by a simple learned temporal weighting. In this section, we show that neither view is sufficient. Instead, the effectiveness of an attack depends on how perturbations propagate through the reverse trajectory and whether their effect survives the subsequent denoising process.

4.1. The failure of sparse peaks and learned heuristics

A natural starting point is to search for a single most vulnerable timestep, or for a sparse subset of important timesteps. To test this, we performed exhaustive single timestep evaluations across extended trajectories, including the full $T = 1000$ process and intermediate bands such as $t \in [200, 600]$. The resulting loss landscape was broadly flat and did not re-

veal a single timestep whose perturbation alone could reliably induce misclassification without also causing substantial perceptual degradation.

We also considered whether temporal importance could be learned through a parametric weighting function. Following the general idea of heuristic interval selection in prior work [5], we parameterised the attack window by a Gaussian distribution $\mathcal{N}(\mu, \sigma^2)$ and defined the temporally weighted adversarial loss as

$$\mathcal{L}_{\text{temporal}} = \sum_{t=1}^T \exp\left(-\frac{(t-\mu)^2}{2\sigma^2}\right) \mathcal{L}_{\text{adv}}(t). \quad (8)$$

In practice, gradient based optimisation did not converge to informative values of μ or σ , and both parameters remained close to their initialisation. This suggests that adversarial vulnerability is not well described by a single dominant timestep or by a simple static temporal weighting. Rather, it emerges from the cumulative interaction between a perturbation and the remaining reverse trajectory.

4.2. From isolated perturbation to sustained steering

The failure of sparse temporal selection highlights an important difference between direct latent perturbation and perturbation in the noise prediction space. A perturbation applied directly to x_t , especially near the final denoising stages, can alter the classifier decision, but it increasingly behaves like a conventional image space attack and often causes visible degradation.

By contrast, perturbing ϵ_θ affects the reverse trajectory more indirectly. An isolated perturbation at a single intermediate timestep may cause a temporary deviation, but the remaining reverse steps can still pull the sample back toward a plausible image configuration. In this sense, isolated perturbations in the noise prediction space are often absorbed by the model’s own generative dynamics. Effective exploitation of this attack surface therefore requires a contiguous temporal window, so that the perturbation can be reinforced across multiple reverse steps rather than introduced only once.

4.3. Propagation energy and the amplification factor

To quantify this behaviour, we measure how a small perturbation introduced at timestep t propagates to the final generated output. Let δ denote a minimal perturbation injected at timestep t , and let $G_t(x_t)$ denote the mapping from the latent state x_t at timestep t to the final generated image x_0 after completing the reverse process. We evaluate two quantities.

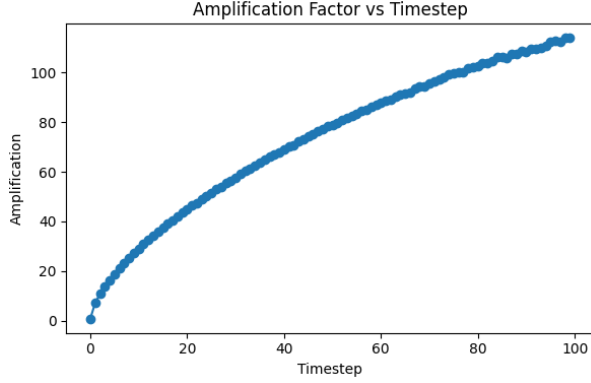


Figure 2. Amplification Factor $\mathcal{A}_t$ measured across the reverse trajectory ($t \in [0, 99]$). The amplification increases monotonically with timestep, indicating growing mathematical sensitivity; however, higher sensitivity does not translate directly into higher adversarial success.

Propagation Energy (E_t). We define the propagation energy as the final output deviation induced by the perturbation:

$$E_t = \|G_t(x_t + \delta) - G_t(x_t)\|_2. \quad (9)$$

Amplification Factor ($\mathcal{A}_t$). We define the amplification factor as the ratio between the output deviation and the size of the initial perturbation:

$$\mathcal{A}_t = \frac{E_t}{\|\delta\|_2}. \quad (10)$$

Evaluating these quantities over the selected interval $t \in [0, 99]$ reveals two broad regimes of reverse diffusion behaviour.

1. **Late stage contractive regime ($t \rightarrow 0$):** At low timesteps, the sample is already close to the image manifold and the reverse process behaves in a relatively stable manner. Perturbations introduced here tend to be attenuated, yielding $\mathcal{A}_t < 1$ in the lowest timesteps. This helps explain why late stage attacks often require larger visible distortion to achieve misclassification.
2. **Early stage high amplification regime ($t \rightarrow 100$):** At high timesteps, the trajectory becomes increasingly sensitive. Even a very small perturbation can produce a large deviation at the final output, resulting in high values of $\mathcal{A}_t$.

Figure 2 shows that the amplification factor increases monotonically with timestep. However, this sensitivity alone does not determine whether an adversarial perturbation remains effective after the rest of the reverse process has unfolded.

4.4. Resolving the Purification Paradox

At first sight, the amplification behaviour appears contradictory. If the trajectory is most sensitive at high timesteps, then one might expect attacks anchored in that region to be the most effective. Our experiments do not support this expectation.

This observation leads to what we term the *Purification Paradox*. A large amplification factor indicates that a perturbation can strongly influence the subsequent trajectory, but it does not guarantee that the induced change will survive as adversarially useful structure. When perturbations are introduced too early, the remaining reverse steps still have substantial capacity to reshape the sample. As a result, the induced deviation may be redirected into plausible image content rather than preserved as an effective adversarial signal.

Empirically, this behaviour is supported by a strong negative correlation between single timestep sensitivity and downstream attack success. In our experiments, the Spearman correlation is approximately $\rho \approx -0.68$, indicating that timesteps with larger local amplification are, in general, less effective starting points for successful attacks. The key implication is that exploiting the reverse diffusion trajectory is not simply a matter of targeting the most sensitive timestep. Instead, the attack window must be anchored in a region where the trajectory remains steerable, yet the remaining denoising process is no longer strong enough to suppress the injected signal. These observations motivate the targeted temporal analysis in the next section, where we identify the interval in which adversarial steering achieves the strongest balance between attack success and perceptual fidelity.

5. Exploiting the Generative Vulnerability Band

Having established that the diffusion trajectory is governed by a Purification Paradox, where maximum sensitivity can trigger semantic restoration, we now examine how this behaviour can be exploited in practice. We evaluate three attack modes, namely x_t , bounded ϵ , and $U\epsilon P$, in order to identify the temporal region in which adversarial steering is most effective and to study the trade off between attack success and perceptual fidelity. This section is intended as a worst case vulnerability analysis of the reverse process. Accordingly, we focus on the temporal region that most clearly exposes adversarial controllability while excluding boundary regimes that are either largely noise dominated or already close to the image manifold.

5.1. Experimental setup and implementation details

To ensure reproducibility, we summarise the main implementation details and hyperparameters used in all experi-

ments.

- **Dataset and evaluation subset:** All experiments are conducted on CIFAR 10. We use a fixed subset of $N_{sub} = 500$ validation images with `data_seed=0`. Adversarial generation is performed with batch size 10 on a single NVIDIA T4 GPU.
- **Diffusion model:** We use a pre trained unconditional DDPM with a backbone trained for $T = 1000$ diffusion steps. For the experiments in this section, the attack is evaluated over a selected reverse trajectory of 100 timesteps, which serves as the attack horizon for the vulnerability analysis.
- **Classifier guided loss:** The adversarial objective is not computed directly on the noisy latent state x_t . Instead, for each attacked timestep we first reconstruct the corresponding clean estimate $\hat{x}_0$ from the predicted noise, that is,

$$\hat{x}_0 = f_{\text{predict}}(x_t, t, \hat{\epsilon}_\theta).$$

The untargeted adversarial loss is then evaluated on $\hat{x}_0$ using a WideResNet 28 10 classifier [9]. In this way, the optimisation acts through the diffusion model while remaining tied to the semantic output presented to the classifier.

- **Optimisation method:** Within each attacked temporal window, we use momentum iterative gradient ascent with $N_{iter} = 20$ optimisation steps and momentum factor 1.0. At each step, the gradient is normalised before the momentum update, and the perturbation is applied using the sign of the accumulated momentum.
- **Perturbation budget:** The perturbation budget is controlled by the scalar parameter ϵ , with the effective ℓ_∞ radius given by $\epsilon/255$. In the experiments reported here, we use $\epsilon = 4$, corresponding to an effective radius of $4/255$.
- **Bounded modes (x_t and bounded ϵ):** The per iteration step size is

$$\alpha = \frac{\epsilon}{N_{iter} \cdot 255}.$$

After each update, the perturbation is projected onto the ℓ_∞ ball of radius $\epsilon/255$ around the initial target.

- **Unrestricted mode ($U\epsilon P$):** The per iteration step size is

$$\alpha = \frac{\epsilon}{255},$$

and no step wise ℓ_∞ projection is applied. As a result, the perturbation in the noise prediction space can accumulate across optimisation steps and across the attacked temporal window.

- **Attack Success Rate (ASR):** ASR denotes the proportion of attacked samples for which the classifier prediction on the attacked reconstruction differs from the ground truth class label. Equivalently, ASR is computed as one minus the adversarial classification accuracy.

5.2. Isolating the Generative Vulnerability Band

To conduct this worst case vulnerability analysis in a controlled manner, we focus on the central part of the reverse trajectory, since evaluating attacks over the full range $t \in [0, 99]$ introduces two confounding regimes:

- **Late stage regime ($t < 20$):** At these timesteps, the sample is already close to the image manifold. Perturbations in this region increasingly resemble conventional image space attacks and therefore reveal less about the generative dynamics of the reverse process.
- **Early stage regime ($t > 80$):** At these timesteps, the latent state remains strongly noise dominated. Gradients in this region are often unstable and less semantically informative.

To avoid these boundary effects, we restrict the temporal analysis to the intermediate interval $t \in [20, 80]$, which we refer to as the *Generative Vulnerability Band*. This interval captures the central part of the reverse process, where semantic structure has begun to emerge, yet the trajectory remains sufficiently plastic for adversarial steering to have a measurable effect.

5.3. High amplification versus low amplification anchoring

Within the Generative Vulnerability Band, we perform a sweep over temporal window sizes $K \in \{5, 10, 15, 20\}$ and compare three anchoring strategies:

- **Top amplification:** windows anchored near $t \rightarrow 80$
- **Low amplification:** windows anchored near $t \rightarrow 20$
- **Random K :** windows placed randomly within the admissible interval

A natural expectation would be that the strongest attacks should arise near the timesteps with the largest local amplification. Our results do not support this view. As shown in Figure 3, perturbations introduced in the high amplification region are often weakened by the remaining reverse dynamics. In other words, local sensitivity alone is not sufficient for persistent adversarial control.

By contrast, windows anchored in the low amplification region are markedly more effective. At this stage of the reverse process, the trajectory has already formed coarse semantic structure, yet the remaining denoising steps do not appear to remove the injected adversarial signal as effectively. Under matched settings with $K = 15$, top amplification anchoring yields only 18.4% white box ASR, whereas low amplification anchoring reaches 96.0% in the x_t mode. This result supports the central observation of the previous section: the most sensitive timesteps are not necessarily the most exploitable ones.

5.4. Attack success and perceptual fidelity

A practically relevant generative attack should not only alter the classifier decision, but should do so while preserving the

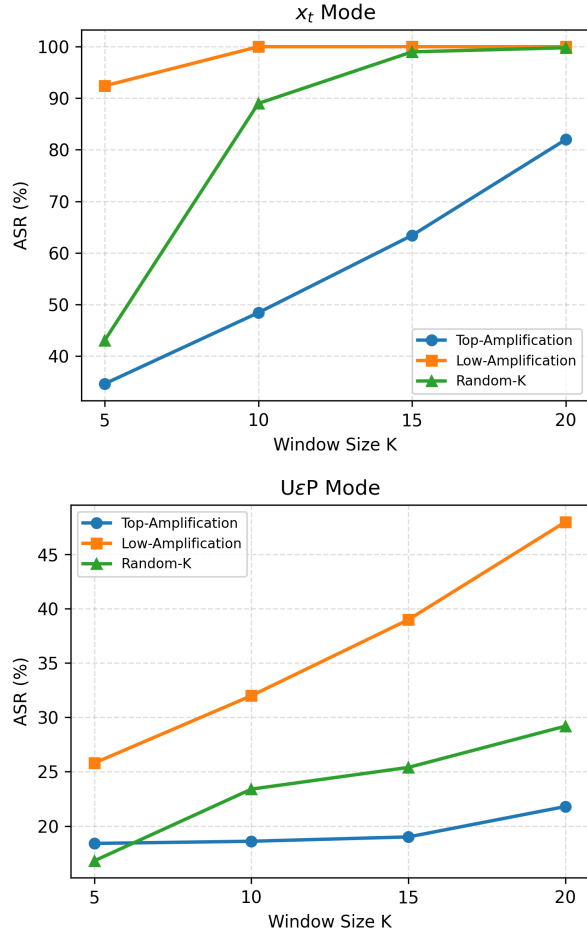


Figure 3. Attack success rate as a function of the temporal window size K under different anchoring strategies. **Top:** x_t mode. **Bottom:** UεP mode. For both attacks, windows anchored in the low amplification regime yield stronger and more persistent adversarial effects than windows anchored near the highest amplification region.

Table 2. Comparison of the three attack modes at $K = 20$.

Mode	ASR (%) $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
x_t mode	100.0	0.640	0.0318
Bounded ϵ mode	18.2	0.770	0.0223
UεP mode	48.0	0.765	0.0219

perceptual plausibility of the generated image. We therefore compare the three attack modes in terms of both attack success rate and visual fidelity, using Structural Similarity (SSIM) [8] and Learned Perceptual Image Patch Similarity (LPIPS) [10].

Table 2 summarises the results at $K = 20$ under low amplification anchoring.

- **x_t mode:** Direct perturbation of the latent state is highly effective in terms of white box attack success, reaching

100.0% ASR. However, this comes at a clear cost in perceptual quality, with SSIM dropping to 0.640 and LPIPS rising to 0.0318.

- **Bounded ϵ mode:** Constraining the perturbation with step wise projection preserves image quality well, giving SSIM 0.770 and LPIPS 0.0223. However, the same constraint substantially limits attack effectiveness, resulting in only 18.2% ASR.
- **UεP mode:** Removing the step wise projection in the noise prediction space improves attack success to 48.0% while maintaining similar perceptual fidelity to the bounded ϵ mode, with SSIM 0.765 and LPIPS 0.0219.

These results show that the unrestricted perturbation in the noise prediction space offers a more favourable balance than the strictly bounded alternative. At the same time, the comparison also makes clear that the direct latent attack on x_t remains the strongest option when attack success is considered alone, although it produces visibly larger distortion.

Figure 4 further illustrates this behaviour across temporal anchoring strategies. The main conclusion is not that local amplification alone determines attack strength, but rather that effective adversarial steering depends on *where* the perturbation is introduced and *how* it is accumulated across the reverse trajectory. In our setting, the most reliable operating region lies in the lower part of the Generative Vulnerability Band, where the reverse process remains steerable without immediately erasing the injected signal.

Overall, the results support two main conclusions. First, the reverse diffusion trajectory contains a temporally localised regime in which adversarial manipulation is substantially more effective than at either boundary of the process. Second, perturbing the noise prediction space without step wise projection provides a stronger balance between attack success and perceptual fidelity than strict bounded optimisation, while remaining less destructive than direct perturbation of the latent state.

6. Semantic Transferability

White box evaluation shows whether an attack can manipulate the source model, but transferability to unseen classifiers provides a stronger test of whether the perturbation captures more general decision relevant structure. We therefore evaluate the adversarial samples generated by UεP on two black box classifiers, namely ResNet56 [2] and MobileNetV2 [6]. Unless otherwise stated, all results in this section use the best setting identified in Section 5, namely a contiguous window of $K = 20$ anchored in the low amplification regime.

Table 3 reveals an interesting pattern. For UεP, the black box ASR exceeds the white box ASR, increasing from 48.0% on the source WideResNet 28 10 model to 67.4% on ResNet56 and 71.4% on MobileNetV2. This suggests that perturbations introduced in the noise prediction space are

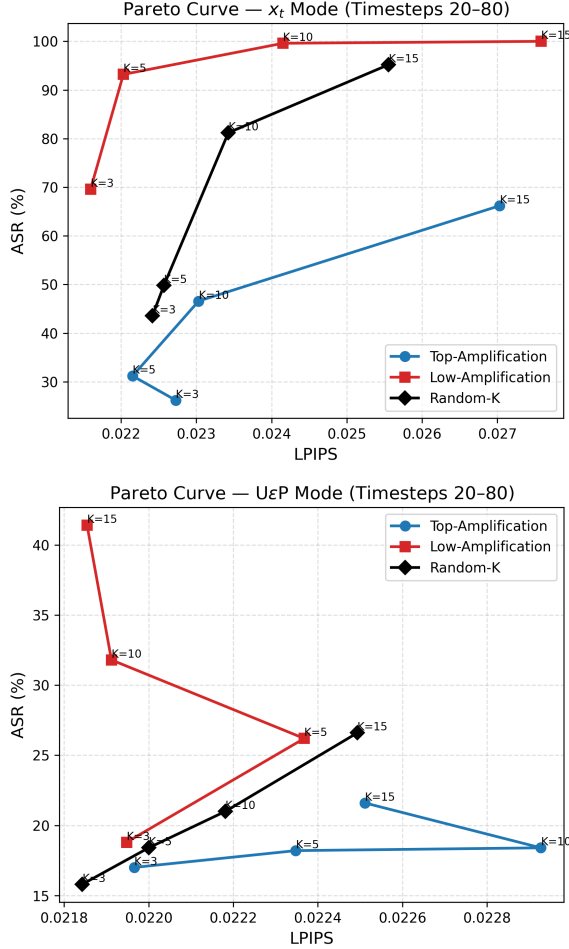


Figure 4. Trade off between attack success rate and LPIPS under different temporal anchoring strategies. The x_t mode achieves high attack success at the cost of rapid degradation in visual quality, whereas UεP yields a smoother trade off, particularly under low amplification anchoring.

Table 3. Transferability evaluation at $K = 20$ with low amplification anchoring. UεP achieves substantially higher black box ASR than bounded ϵ mode while maintaining similar perceptual quality.

Attack Mode	Source ASR (%)	ResNet56 ASR (%)	MobileNetV2 ASR (%)	SSIM
x_t Mode	100.0	88.6	86.0	0.640
Bounded ϵ Mode	18.2	57.8	66.6	0.770
UεP Mode	48.0	67.4	71.4	0.765

not merely overfitted to the source classifier, but can also alter features that transfer across unseen architectures.

The comparison with the other attack modes helps interpret this result. Direct perturbation of x_t transfers strongly across models, but does so with a clear loss of perceptual quality, as reflected by the much lower SSIM of 0.640. By contrast, bounded ϵ mode preserves image quality but remains relatively weak in the white box setting. UεP oc-

cupies an intermediate position: it preserves perceptual structure much better than direct latent perturbation while achieving substantially stronger black box transfer than the bounded variant.

A plausible interpretation is that direct perturbation of x_t mainly induces broad visual corruption, which disrupts many classifiers but is also perceptually obvious. In contrast, UεP perturbs the reverse trajectory through the predicted noise and therefore alters classifier relevant structure in a more distributed manner. This may explain why its transferability remains strong even when its white box ASR is lower than that of the x_t mode. Overall, these results suggest that adversarial steering in the noise prediction space can produce perturbations that are both more transferable than bounded latent consistent attacks and substantially less destructive than direct perturbation of x_t .

7. Discussion and Conclusion

We analysed adversarial vulnerability along the reverse diffusion trajectory and identified a phenomenon that we term the *Purification Paradox*: timesteps with high local sensitivity are not necessarily the most effective attack points, because subsequent denoising steps can suppress or absorb the perturbation. By shifting the attack surface from direct latent perturbation (x_t) to perturbation of the noise prediction (ϵ_θ), we showed that effective attacks depend not only on local sensitivity, but also on whether the injected signal survives the remaining reverse process. Motivated by this observation, we introduced Unrestricted ϵ -Space Perturbation (UεP) and showed that attacks are most effective within an intermediate Generative Vulnerability Band ($t \in [20, 80]$). In this regime, UεP achieves a stronger balance between attack success and perceptual fidelity than direct latent perturbation, while also transferring effectively across unseen classifiers, reaching 71.4% ASR on MobileNetV2 with SSIM above 0.76. These results suggest that diffusion robustness should be assessed at the trajectory level rather than only at the final image level. Since UεP can produce perceptually plausible yet semantically shifted outputs, methods based only on visible artefacts may be insufficient. Although our study is limited to DDPMs and CIFAR 10, the same trajectory level analysis may help characterise vulnerability in larger diffusion models and inform the development of more reliable defences.

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